

WEEK 7 PROJECT SUMMARY

HealthConnect Experience Lab — Data Analytics Track | HC-POD 8

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1. What was planned to be tested

Week 7 was planned as an end-to-end validation of the Week 6 analytical outputs and Power BI dashboard. The planned testing covered KPI calculations, important analytical findings, segment definitions, dashboard visuals, slicers, DAX calculations, Lead Time Band ordering, and cross-track dependencies.

2. What was actually tested

All 5 KPI cards were compared with independent calculations.

All 6 comparison visuals were checked against independently calculated results.

Appointment Type and Gender slicers were tested.

The combined Lead Time $\times$ Previous No-Show visual was tested with slicer selections.

Lead Time Band ordering was tested and the underlying sort configuration was corrected.

The 64.1% Data Analytics versus 70.5% Data Science segment-rate discrepancy was tested collaboratively.

3. Most important testing results

The five KPI cards and six comparison visuals matched the independent calculations. The two slicers responded correctly to selections and clearing. The combined Lead Time $\times$ Previous No-Show calculations also matched independent results across the tested filters. The Lead Time Band ordering was corrected and subsequently re-tested successfully.

4. Issues or weaknesses identified

A Lead Time Band sort-order issue was identified in the Power BI dashboard.

The 64.1% versus 70.5% cross-track segment-rate discrepancy required investigation.

A small-sample reversed pattern was observed in the 'Prefer not to say' gender group and was retained as a caution rather than a general conclusion.

The analysis remains limited by the synthetic dataset and observational nature of the findings.

5. Improvements/refinements made

A separate Lead Time Band Sort column was created directly from `booking_lead_days` to resolve the previous circular dependency and establish the intended chronological ordering: Same week (0–7 days), 2–3 weeks (8–21 days), and 3+ weeks (22–60 days). Analytical documentation was also refined to state segment thresholds and denominator definitions explicitly.

6. Retesting results

After the refinement, the 5 KPI cards, 6 comparison visuals, Appointment Type slicer, Gender slicer, combined Lead Time $\times$ Previous No-Show visual, and Lead Time Band ordering were re-tested. All tested components continued to function correctly. No new calculation or filtering errors were identified. Overall re-test result: PASS.

7. Track(s) collaborated with

Data Analytics collaborated with the Data Science Track, specifically Donald Nwachukwu, for formal cross-track testing.

8. What was tested collaboratively

The collaborative test focused on the Lead Time $\times$ Previous No-Show segment-rate discrepancy. Donald provided the exact Data Science definition: `booking_lead_days > 30` and `previous_no_shows  $\geq$  1`. The Data Analytics track independently re-ran this definition using the same underlying dataset and tested both denominator variants.

9. What changed as a result

The cross-track testing showed that the Data Science results could be independently reproduced: 70.52% when Cancelled appointments were excluded and 67.88% when they were included. The investigation clarified that the original analyses used different lead-time thresholds and denominator definitions. Future comparisons will explicitly state these criteria.

10. Key findings or validation outcomes

Overall No-Show Rate was validated at 48.5%.

Booking lead time remained an important observed factor: 27.8% same week, 37.2% at 2–3 weeks, and 56.9% at 3+ weeks.

Previous No-Show history remained higher at 55.4% versus 43.5% with no previous No-Show history.

The combined Data Analytics segmentation reached a highest observed rate of 64.1% for 3+ weeks plus previous No-Show history.

The combined six-category analysis had Cramér's $V = 0.189$.

The dashboard's tested calculations and filtering behaviour were validated.

11. Major challenges encountered

The main challenges were resolving the Lead Time Band sorting configuration and tracing the difference between the Data Analytics and Data Science segment rates. The latter required checking the exact segment threshold and denominator rather than assuming the discrepancy was caused by only one factor.

12. Important decisions made

Keep the validated Week 6 dashboard structure rather than rebuilding it.

Correct the Lead Time Band sorting using a separate sort column.

Treat the 64.1% and 70.5% figures as results from different definitions unless their criteria are explicitly aligned.

Use cautious language for observational findings and avoid causal claims.

Document successful validations as well as identified issues instead of assuming errors.

13. Remaining limitations

The HealthConnect dataset is synthetic.

Previous appointment and No-Show history may not represent fully coherent longitudinal patient histories.

Missing values remain in reminder_channel, distance_to_clinic_km, and waiting_time_minutes.

The observational analysis does not establish causation, including reminder effectiveness.

Some smaller subgroups require cautious interpretation.

14. Remaining issues or dependencies

No confirmed KPI calculation errors remain in the tested dashboard configuration. Remaining considerations include maintaining consistent segment definitions across tracks, further evaluating predictive usefulness of the combined segmentation, and reassessing findings if more representative operational data becomes available.

15. Contribution to the overall HealthConnect project

The Data Analytics contribution provided validated appointment No-Show analysis, KPI validation, segment testing, dashboard validation and refinement, evidence-based business insights and recommendations, and cross-track testing with Data Science. The Week 7 work strengthened the reliability and interpretability of the HealthConnect analytical output and established a validated dashboard baseline for continued integration.

16. What must be completed before Week 8

Carry forward the validated analytical findings and dashboard baseline.

Maintain explicit definitions for segment thresholds, No-Show conditions, and denominator rules.

Continue cross-track integration where analytical outputs become dependencies.

Further evaluate the predictive usefulness of the validated segments.

Reassess findings when more representative data becomes available.

Use the validated Week 7 outputs as the starting point for the next stage rather than restarting the analysis.